

# Advanced Computing Project

Projeto em Computação Avançada (MEI/MCA)

## Tutorial 1: How To Search and Analyze Related Work?

University of Minho  
2025-2026



# Why related work matters

## Motivation

- ❖ Avoid reinventing the wheel
  - There is a good chance someone has already thought of the same solution as you
  - All papers have related work, unless you propose a new research field!
- ❖ Understand the state-of-the-art
  - Identify gaps/limitations
- ❖ Justify the novelty of your work
  - Position your contribution with existing work

# Why related work matters

## Common mistakes

- ❖ *“Because my advisor said so!”*
- ❖ Searching randomly
- ❖ Early stopping/checking the top results
- ❖ Using AI to understand novelty
- ❖ Not understanding how papers connect

# From idea to search strategy

❖ “*I want to make AI more energy-efficient*”

► **Search:** “*Energy efficiency in AI*”

About 6 730 000 results (0,19 sec)

## AI and energy efficiency

[R Omar](#) - 2023 IEEE 20th International Conference on ..., 2023 - [ieeexplore.ieee.org](#)

... Most of the studies are focusing on increased accuracy rather than **energy efficiency** ... **energy** cost associated with **AI** models, and techniques to help increase the **energy efficiency** of **AI** ...

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## [PDF] Enhancing **energy efficiency** with **ai**: a review of machine learning models in electricity demand forecasting

[AO Aderibigbe](#), [EC Ani](#), [PE Ohenhen...](#) - Engineering Science & ..., 2023 - [academia.edu](#)

... of **artificial intelligence (AI)** and machine learning (ML) on enhancing **energy efficiency**, ... by the emergence of **AI** in **energy efficiency**, focusing on the role of **AI** in electricity demand ...

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## [HTML] **Artificial intelligence** evolution in smart buildings for **energy efficiency**

[H Farzaneh](#), [L Malehmirchegini](#), [A Bejan](#), [T Afolabi...](#) - Applied Sciences, 2021 - [mdpi.com](#)

... can offer a possible solution to this problem by providing **energy efficiency**, comfort, health ... of **AI** and **AI**-based approaches in smart buildings, especially for building **energy efficiency**. ...

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## Green **AI**: Enhancing sustainability and **energy efficiency** in **AI**-integrated enterprise systems

[S Dash](#) - IEEE Access, 2025 - [ieeexplore.ieee.org](#)

... MONITORING AND MEASUREMENT The primary goal of this study is to make **AI** model training and inference **energy efficient**. That is why it is essential to track the **energy** consumption...

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# From idea to search strategy

❖ “*I want to make AI more energy-efficient*”

▸ **Search:** “*Energy efficiency in AI*”

❖ We should start with descriptive keywords

▸ The same words might mean different things for different people

❖ Use synonyms and specific use cases



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**AI and energy efficiency**  
[R Omar](#) - 2023 IEEE 20th International Conference on ..., 2023 - [ieeexplore.ieee.org](#)  
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# From idea to search strategy

GPU power management for AI training

Q

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## Benchmarking the performance and **energy** efficiency of **AI** accelerators for **AI training**

[Y Wang](#), [Q Wang](#), [S Shi](#), [X He](#), [Z Tang](#)... - 2020 20th IEEE/ACM ..., 2020 - [ieeexplore.ieee.org](#)

... processors (ie, Intel CPU, **NVIDIA GPU**, AMD **GPU**, and Google TPU) in **training** DNNs, we carry out a ... 4.3 The **power management** of the AMD **GPU** is inferior to that of the **NVIDIA GPU**, ...

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## Single-node **power** demand during **ai training**: Measurements on an 8-gpu **nvidia** h100 system

[I Latif](#), [AC Newkirk](#), [MR Carbone](#), [A Munir](#), [Y Lin](#)... - IEEE ..., 2025 - [ieeexplore.ieee.org](#)

... time, potentially enabling finer-grained **energy management** without compromising model quality. These hardware-software co-optimization approaches could collectively yield ...

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## Towards improved **power management** in cloud **gpus**

[P Patel](#), [Z Gong](#), [S Rizvi](#), [E Choukse](#)... - IEEE Computer ..., 2023 - [ieeexplore.ieee.org](#)

... To show this behaviour, we run spmv and vgg16-**training** separately on A100 with 2 MIG partitions, at each workload's lowest feasible SM frequency under a 3% bound on performance ...

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## **Power** stabilization for **AI training** datacenters

[E Choukse](#), [B Warriar](#), [S Heath](#), [L Belmont](#)... - arXiv preprint arXiv ..., 2025 - [arxiv.org](#)

... **Artificial Intelligence (AI) training** workloads spanning several tens of thousands of **GPUs** present unique **power management** ... variability in **power** consumption during the **training**. Given ...

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GPU power management for AI training

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optimizing GPU energy consumption of AI training

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[Y Wang](#), [Q Wang](#), [S Shi](#), [X He](#), [Z Tang](#)... - 2020 20th IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGRID), 2020 - [ieeexplore.ieee.org](#)

... , **NVIDIA GPU**, AMD **GPU**, and Google TPU) in **training** DNNs, we carry out a comprehensive empirical study on the performance and **energy efficiency** ... -level **optimization** of better **GPU** ...

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Dynamic **GPU energy optimization** for machine **learning training** workloads

[F Wang](#), [W Zhang](#), [S Lai](#), [M Hao](#)... - IEEE Transactions on Cloud Computing, 2021 - [ieeexplore.ieee.org](#)

... find a trade-off between execution time and **energy consumption**. We evaluate the GPOEO by applying it to 71 machine **learning** workloads from two **AI** benchmark suites running on an ...

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Green **ai**: A preliminary empirical study on **energy consumption** in dl models across different runtime infrastructures

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... the **energy usage** of **training AI** models. Fischer et al. [5] also investigated the **energy efficiency** ... container incorporates a validated set of libraries designed to enable and **optimize GPU** ...

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[J You](#), [JW Chung](#), [M Chowdhury](#) - 20th USENIX Symposium on Operating Systems, 2023 - [usenix.org](#)

... Benchmarking the performance and **energy efficiency** of **AI** accelerators for **AI training**. In 20th IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGRID), ...

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[E Choukse](#), [B Warrior](#), [S Heath](#), [L Belmont](#)... - arXiv preprint arXiv:2501.00001, 2025 - arxiv.org

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reducing energy in large model training

About 4 280 000 results (0,43 sec)

Reducing **energy** bloat in large model training

[JW Chung](#), [Y Gu](#), [I Jang](#), [L Meng](#), [N Bansal](#)... - Proceedings of the ACM Conference on AI and ML, 2024 - dl.acm.org

... **energy**-efficiency gains [15, 17, 78, 85, 86]. In this paper, we seek a software method ... **reduces** the **energy** consumption of **large model training** without slowdown, thereby also **reducing** ...

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Minimizing **energy** consumption of deep learning models by **energy-aware training**

[D Lazzaro](#), [AE Cinà](#), [M Pintor](#), [A Demontis](#)... - ... Conference on Image and Vision Computing, 2023 - Springer

... We can observe a slight variation in the **energy** ratio when considering **large** values for  $\sigma$ . This effect is due to the approximation function  $\hat{\ell}_0$  in Eq. 2 not being ...

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E2-train: **Training** state-of-the-art cnns with over 80% **energy** savings

[Y Wang](#), [Z Jiang](#), [X Chen](#), [P Xu](#)... - Advances in neural network information processing, 2019 - proceedings.neurips.cc

... is to **reduce** the total **energy** cost in **training**, which ... **training** in **large-scale** GPU clusters. In contrast, our focus is to trim down the total **energy** cost for in-situ, resource-constrained **training**...

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Green/Energy-Efficient Ai: Strategies for **Reducing** Power Usage in **Large-Scale Model Training**

[A Garg](#), [A Bhosale](#) - 2025 IEEE International Conference on AI and ML, 2025 - ieeexplore.ieee.org

... as a guide for applying green **training** strategies to **large-scale** AI development. Thanks to a ... workable framework aimed at **reducing energy** consumption while maintaining performance...

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# Where to search

## ❖ Academic search engines

- Google scholar
- DBLP (computer science)
- Semantic scholar
- Research gate
- Connected papers

## ❖ Publisher platforms

- ACM Digital Library
- IEEE Xplore
- USENIX
- ArXiv
- And many more!

## ❖ AI agents and platforms

- ChatGPT
- Perplexity
- Semantic scholar + AI summaries
- Elicit
- Consensus
- Scite
- ...

# Where to search

## AI agents and platforms

### ❖ Use for:

- Discovery
- Summarization
- Clustering
- Pattern detection

### ❖ Do not use for:

- Replace deep reading
- Validate experimental rigor
- Understand (hidden) assumptions/insight
- Replace your reasoning

# Where to search

## Searching by citations: forward and backward referencing

DOI: 10.48550/arXiv.2208.06102 • Corpus ID: 251554526

### Zeus: Understanding and Optimizing GPU Energy Consumption of DNN Training

[Jie You](#), [Jaehoon Chung](#), [Mosharaf Chowdhury](#) • Published in [Symposium on Networked...](#) 12 August 2022 •

Computer Science, Engineering

**TLDR** Zeus, an optimization framework to navigate the tradeoff between energy consumption and performance optimization by automatically finding optimal job- and GPU-level configurations for recurring DNN training jobs, is proposed. [Expand](#)



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# Where to search

## Searching by citations: forward and backward referencing

130 Citations

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Energy-Efficient GPU Clusters Scheduling for Deep Learning

Diandian Gu

Xintong Xie

Gang Huang

Xin Jin

Xuanzhe Liu

Computer Science, Engineering · [arXiv.org](#) · 2023

TLDR

PowerFlow is proposed, a GPU clusters scheduler that reduces the average Job Completion Time (JCT) under an energy budget and applies network packing and buddy allocation to job placement, thus avoiding extra energy consumed by cluster fragmentations.[Expand](#)

👍 10

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[PDF]

≡ 4 Excerpts

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EnvPipe: Performance-preserving DNN Training Framework for Saving Energy

Sang-Jun Choi

Inhoe Koo

Jeongseob Ahn

Myeongjae Jeon

Youngjin Kwon

Computer Science, Engineering · [USENIX Annual Technical Conference](#) · 2023

TLDR

E NV P IPE aims to maximize energy saving while maintaining negligible performance slowdown, and is implemented as a library using PyTorch and can save up to 25.2% energy in single-node training with 4 GPUs and 28.4% in multi- node training with 16 GPUs, while keeping performance degradation to less than 1%.[Expand](#)

👍 27

💡 Highly Influenced

PDF

≡ 4 Excerpts

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PowerTrain: Fast, generalizable time and power prediction models to optimize DNN training on accelerated edges

## Which works have cited this paper?

# Where to search

## Searching by citations: forward and backward referencing

**104 References**

 Date Range ▾ Citation Type ▾ Has PDF Author ▾ More Filters Sort by Relevance ▾

### The Impact of GPU DVFS on the Energy and Performance of Deep Learning: an Empirical Study

Zhenheng Tang Yuxin Wang Qiang Wang X. Chu Computer Science, Engineering · [Energy-Efficient Computing and Networking](#) · 2019

**TLDR** This work empirically investigates the impact of GPU Dynamic Voltage and Frequency Scaling (DVFS) on the energy consumption and performance of deep learning and suggests that GPU DVFS has great potentials to help develop energy efficient DNN training/inference schemes. [Expand](#)

 113  [PDF]  2 Excerpts  Save

### Towards Power Efficiency in Deep Learning on Data Center Hardware

M. Hodak Masha Gorkovenko A. Dholakia Computer Science, Engineering · 2019 IEEE International Conference on Big Data... · 2019

**TLDR** This work directly measure power used by the whole system as well as that used by GPU, CPU, and RAM during DL training to determine their contributions to the overall energy consumption, finding that while GPUs use most of the power, the consumption of other components is also significant and their optimizations can bring important power savings. [Expand](#)

 33  1 Excerpt  Save

[Dynamic GPU Energy Optimization for Machine Learning Training Workloads](#)

## Which prior work does this paper cite?

# Where to search

## Searching by citations: forward and backward referencing

DOI: 10.1145/359545.359563 • Corpus ID: 215822405

### Time, clocks, and the ordering of events in a distributed system

L. Lamport • Published in CACM 1 July 1978 • Computer Science • Commun. ACM

TLDR

A distributed algorithm is given for synchronizing a system of logical clocks which can be used to totally order the events, and a bound is derived on how far out of synchrony the clocks can become.[Expand](#)

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ADVANCED COMPUTING PROJECT

TUTORIAL 1: HOW TO SEARCH AND ANALYZE RELATED WORK?

11

# Where to search

## Searching by conference

❖ **General systems research:** SOSP, OSDI, EuroSys, ATC, Middleware, SoCC

❖ **Specific domains:**

- **HPC:** Supercomputing, HPDC, IPDPS, ICS, CCGrid, ICPP, Cluster
- **Storage:** FAST, Systor, HotStorage
- **Network:** NSDI, SIGCOMM, INFOCOM
- **Databases:** VLDB, SIGMOD, ICDE
- **Computer Architecture:** ASPLOS, MICRO, HPCA, ISCA
- **AI\*:** NeurIPS, ICML, ICLR, AAAI

# Where to search

## Searching by author



Jie You

University of Michigan  
Verified email at umich.edu

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| TITLE                                                                                                                                                                                                                                                   | CITED BY | YEAR |
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| <a href="#">Near optimal coflow scheduling in networks</a><br>M Chowdhury, S Khuller, M Purohit, S Yang, J You<br>The 31st ACM Symposium on Parallelism in Algorithms and Architectures, 123-134                                                        | 34       | 2019 |
| <a href="#">{AdaEmbed}: Adaptive embedding for {Large-Scale} recommendation models</a><br>F Lai, W Zhang, R Liu, W Tsai, X Wei, Y Hu, S Devkota, J Huang, J Park, ...<br>17th USENIX Symposium on Operating Systems Design and Implementation (OSDI ... | 21       | 2023 |
| <a href="#">Client-side name collision vulnerability in the new gtld era: A systematic study</a><br>QA Chen, M Thomas, E Osterweil, Y Cao, J You, ZM Mao<br>Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications ...            | 18       | 2017 |

# Where to search

## Searching by author



Jie You

University of Michigan  
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Jae-Won Chung

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| <a href="#">Humanity's last exam</a><br>L Phan, A Gatti, Z Han, N Li, J Hu, H Zhang, CBC Zhang, M Shaaban, ...<br>arXiv preprint arXiv:2501.14249                                                                         | 355      | 2025 |
| <a href="#">Zeus: Understanding and Optimizing GPU Energy Consumption of DNN Training</a><br>J You*, JW Chung*, M Chowdhury<br>USENIX NSDI                                                                                | 187      | 2023 |
| <a href="#">Andes: Defining and Enhancing Quality-of-Experience in LLM-Based Text Streaming Services</a><br>J Liu, JW Chung, Z Wu, F Lai, M Lee, M Chowdhury<br>arXiv preprint arXiv:2404.16283                           | 48       | 2024 |
| <a href="#">Reducing Energy Bloat in Large Model Training</a><br>JW Chung, Y Gu, I Jang, L Meng, N Bansal, M Chowdhury<br>ACM SOSP                                                                                        | 35 *     | 2024 |
| <a href="#">The ML. ENERGY Benchmark: Toward Automated Inference Energy Measurement and Optimization</a><br>JW Chung, J Liu, JJ Ma, R Wu, OJ Kweon, Y Xia, Z Wu, M Chowdhury<br>NeurIPS Datasets & Benchmarks (Spotlight) | 12       | 2025 |
| <a href="#">Toward Cross-Layer Energy Optimizations in AI Systems</a><br>JW Chung, N Talati, M Chowdhury<br>DoE ASCR Energy Efficient Computing for Science Workshop                                                      | 6        | 2024 |

# How to filter papers

## Three-pass reading strategy

### ❖ Pass 1 (5 minutes):

- Title
- Abstract
- Figures
- Contributions list

**Pass 1 and 2:** select/filter key papers

**Pass 3:** understand the papers

# How to filter papers

## Three-pass reading strategy

### ❖ Pass 1 (5 minutes):

- Title
- Abstract
- Figures
- Contributions list

### ❖ Pass 2 (20-30 minutes):

- Introduction
- Challenges
- (Key) Design
- Evaluation methodology

**Pass 1 and 2:** select/filter key papers

**Pass 3:** understand the papers

# How to filter papers

## Three-pass reading strategy

### ❖ Pass 1 (5 minutes):

- Title
- Abstract
- Figures
- Contributions list

### ❖ Pass 2 (20-30 minutes):

- Introduction
- Challenges
- (Key) Design
- Evaluation methodology

### ❖ Pass 3 ( $\infty$ minutes):

- Deep dive
- Technical details
- Key insight and assumptions
- Experimental setup and result analysis

**Pass 1 and 2:** select/filter key papers

**Pass 3:** understand the papers

# How to analyze related work

Problem and solution

# How to analyze related work

## Problem and solution

### ❖ Understanding the problem

- Does the problem addressed in the paper match yours?
- Does it have the same assumptions? If not, do they make sense to your scenario (and should you consider them)?
- **Relevant sections:** introduction, background, motivation/experimental study, related work

# How to analyze related work

## Problem and solution

### ❖ Understanding the problem

- Does the problem addressed in the paper match yours?
- Does it have the same assumptions? If not, do they make sense to your scenario (and should you consider them)?
- **Relevant sections:** introduction, background, motivation/experimental study, related work

### ❖ Understanding the solution

- What is the main goal and insight (secret sauce!)?
- What are the key contributions?
- How radically different is the solution from prior work?
- **Relevant sections:** introduction, design, experimental evaluation

# How to analyze related work

## Identifying gaps

- ❖ Unrealistic assumptions
- ❖ Ignored scenarios
- ❖ Poor experimental evaluation
- ❖ Hardware assumptions
- ❖ Overlooked workloads

# How to analyze related work

Identifying patterns

# How to analyze related work

## Identifying patterns

### ❖ Identify key features common to all papers

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- Solution/techniques
- Hardware
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### ❖ Positioning your work

- How does your solution compare to others (what is new)?
- What scenarios does your solution take into account that others have overlooked?

# Questions?